

List of main modifications done to AnimalTA for the update v4.2.0

Video reader

- The video reader will now display the coordinates of the cursor when moving over the video (just above the play/pause buttons)

Video edition

- Video conversion will be faster and more robust to corrupted files.
- It is now possible to choose the quality reduction strength (trade-off between video quality and video weight/rapidity of the video tracking). Advised value is 5.
- Two videos with the same name can now be loaded, the program will propose to rename the new one or rewrite the old one.

Video preparation

- When changing arenas, background, cropping, scale, stabilization, parameters of the tracking: after validation, the last modified video is selected.
- The background panel now allows you to validate and go to the next video. Only videos with a background prepared will be available.
- It is now possible to rename a video after it has been tracked by clicking on the cross on the right of the video name in the main table.
- In the cropping section, it is now possible to select both vertical and horizontal separators for spatial cropping at the same time.
- To facilitate the cropping, it is now possible to draw the arena first on the video (section define arenas) and to click the “More options > Adapt cropping” so that the spatial cropping is automatically done around the arena.
- In video preparation, when using an adaptive thresholding, it is now possible to change the sensibility of the thresholding playing with a second parameter.
- In video preparation, when using a dynamical background, it is now possible to change the window length used to build the changing background (using the “P” button).

Importation

- It is now possible to import data from other video-tracking programs, allowing to correct tracking obtained by other means than AnimalTA.

Tracking

- Faster than before: the program will now use multiprocessing.

- It is now possible to track the head/tail of the animals (for elongated animals) and not only the center of the body. To use this option, select the “Separate head from tail” option in the “begin tracking” panel.
- The “correct flickering” option has been removed.

View and correct tracking

- Possibility to remove all the data points in a particular area of the video. Press the “e” key on the keyboard to unlock this option. Once it is done, selecting a part of the video using “Ctrl + click and drag” will remove all the data points inside the selection. Press again “e” to deactivate this option. If the option is activated, you will see the symbol [X] in the bottom right corner of the video.

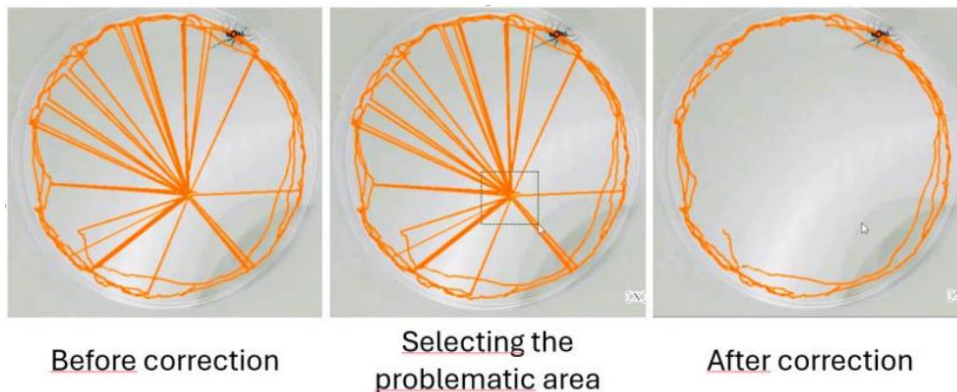


Illustration of removing data in a specific area of the video.

- The "N" shortcut will jump to the next missing frame
- Addition of a "next conflict" button which will jump to the next frame for which the speed was abnormally elevated considering the movement pattern of the target
- Addition of a “jump to next non-missing value” button.
- Using "Alt" key + left click will move the selected individual coordinates' to where the click occurs.
- Improved resolution of the tracking data showed on the screen.
- Shift + arrow when renaming a target allows to select characters.
- When using the jump to next missing/non-missing data, the corresponding individual is selected.
- Supplementary options are available from a new top-menu above the video.
- Possibility to click on the 1st row, 1st column of the coordinates table to look for a specific target name in the table.

Analysis

- If the videos are very long, the analysis part won't calculate all the data but make an approximation for the preview, this will prevent the program from being too

slow with long videos (a red warning will then be displayed to inform user that this is only preview results).

Exploration

- Possibility to save heatmaps obtained from rectangular/circular meshes.

Sequences

- Possibility to use the "Auto-range" technique that will cut the video into regular segments for the analysis. These cuts may be based on Time (for instance will separate a 10min videos in 1min sequences); on Exploration; on interactions with elements of interest (each time the animal is inside/outside an element of interest).
- Possibility to use the event "cross a segment" to separate the video into sequences.
- Possibility to remove sequences in multiple arenas/videos.

More analysis

A new "more analysis" button is now available, allowing to add some supplementary and facultative analysis.

- Save detailed data regarding segments crosses: It will save every event of segment crossing (who crossed and when).
- Durations of movements and stops: The program will calculate the duration of each stop/move event and will either calculate its mean duration (Mean_dur_stops/moves) and add these values to the "Results_by_ind" file or save the data related to every single event (Events_dur_stops/moves needed for survival analysis). In that second case a new file will be created in the Results section with that data (file name: Stops or Moves). The file will also indicate whether the stop/move event was censored. Censored data are coded with a 1 value while non censored are coded 0. Censored events are either stops/moves beginning at the first or ending at the last frame of the Sequence, or moves/stops interrupted by NA values. Another parameter can be defined to ignore small events interrupting a stop/movement, the maximum duration of these interrupting events to be neglected can be chosen.
- Detailed data columns: It is now possible to choose which columns should be saved in the "Detailed data" files. New optional columns are: "Frame", "Acceleration",

“Orientation”, “Angle change”, “Angular speed”, “Meander”.

Export

- Possible to add a black mask on the part of the videos that are not part of an arena
- Possible to draw the trajectories on top of a white background
- Possibility to export an image of the video treatments/trajectories.

Bug fixes

- Using both video rotation and stabilization on the same video was creating a bug and making the program fail the tracking. This bug is now fixed.
- Distance to segments was not properly saved in the Detailed data files. This is now fixed.
- In small videos, the arenas shapes were sometimes filled by the color, this is fixed
- In the background panel, the button "Use the first frame" was not working, it is fixed
- In the “doing arenas” option: Arenas are now easier to select and won't be moving together unless if they are from the same shape
- Stabilisation was not properly applied when doing the "redo tracking" option. It is fixed
- Some videos were deformed after conversion, this should not happen anymore